Linköping Studies in Science and Technology
Licentiate Thesis No. 2001

On Early Product Development in the Context of Aircraft Thermal Management Design

Adelia Darlene Drego

LINKÖPING UNIVERSITY

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This is a Swedish Licentiate's Thesis
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Abstract

Military aircraft face several contemporary challenges. From a thermal management perspective, they include the more extensive use of electrically driven technologies and more power-hungry tactical systems. These systems demand more cooling power from the aircraft thermal management system. Thermal management systems of modern aircraft face another challenge with shrinking heat sink capability. This is because of platform design aspects. Increasing use of composite materials for airframe skin over traditional metals impedes the dissipation of waste heat through skin. Infrared and radar cross-section signatures are minimised to improve aircraft stealth by minimising the cross-sectional areas of ram air intakes. This reduces the amount of waste heat that can be dumped over-board through ram air. And there are only two options for heat sinks, air (ram or engine fan) and fuel. All these contemporary challenges create a strong need to carry out thermal management design effectively at the aircraft concept stage.

A framework is presented in this thesis that demonstrates how effective thermal management design at the concept stage can be conducted at an aircraft developer like Saab. The framework was created with data collection through workshops, document studies, interviews, and group discussions conducted at Saab. Therefore, the framework is based on industrial reality. It can be tested for application at the aircraft concept stage of Saab projects. Further, the methods created in this thesis can be used in a broader context that transcend their primary application in thermal management design. They can also complement other methods presented in literature on aircraft thermal management. Thus, this thesis makes industrial and scientific contributions to aircraft thermal management design.

Preface

The saying goes "*It takes a village*….". It certainly has taken a village for me to get to this milestone.

Firstly, I would like to thank my supervisory team. To my main supervisor, Hampus Gavel for your strong encouragement and support through thick and thin. And, for always being a sounding board for my research challenges. To Anton Wiberg and Ingo Staack, for helping me realise my 'napkin' ideas into research studies. A special thanks to Anton for always, always just being there. And, finally to Johan Ölvander for always helping me see the big picture.

To my colleagues at the 'Vehicle Systems Simulation and Concept Development' Section at Saab Aeronautics, if I have seen further, it is by standing on the shoulders of giants! Thank-you! I have learnt so much from you in the last four years. A special thanks to my section manager, Peter Gotenstam for always supporting my research ideas and work.

To my PhD and post-doc colleagues at the Division of Product Realisation at Linköping University, for all the laughs and conversations at the big fika table. Thank you for always making me feel like part of the PROD family.

And, finally to my loved ones, for your love, support, and encouragement over the years. And, for always keeping my feet firmly planted on the ground while I look up at the sky.

Adelia Darlene Drego
Tannefors, Linköping, July 2024

Nomenclature

ChatGPT Chat generative pre-trained transformer

DRM design research methodology

DSM design/dependency structure matrix

ECS environmental control system

KFS (in Swedish) konceptutveckling för framtida stridsflygplan

PDP product development process

RQ research question

TM thermal management

TMS thermal management system

VCS vapor cycle system

List of Appended Papers

I. Drego, Adelia, "Harnessing the Complexity for Vehicle System Design at the Concept Design Phase of an Aircraft," *Proceedings of the Design Society*, Vol.2, 2022, pp. 1845-1854, DOI: https://doi.org/10.1017/pds.2022.187

II. Drego, Adelia; Steinkellner, Sören, "A System Integrator's Perspective on Vehicle System Evaluation at the Aircraft Concept Stage," *33rd Congress of the International Council of the Aeronautical Sciences*, 2022, DOI: https://www.icas.org/ICAS_ARCHIVE/ICAS2022/data/papers/ICAS2022_0856_paper.pdf

III. Drego, Adelia; Andersson, Daniel; Staack, Ingo, "Parameter Tuning of a Vapor Cycle System for a Surveillance Aircraft," *MDPI Aerospace,* Vol. 11, No. 1, 2024, DOI: https://doi.org/10.3390/aerospace11010066

IV. Drego, Adelia; Wiberg, Anton; Staack, Ingo, "Cool Planes for Hot Missions: Early but Effective Aircraft Thermal Management Design,"
Manuscript accepted for presentation at The American Institute of Aeronautics and Astronautics Aviation Forum 2024.

V. Drego, Adelia; Wiberg, Anton; Gavel, Hampus, "Chilling with ChatGPT: Can Future Fighter Aircraft Environmental Control Systems be Crafted with AI Assistance?"
Manuscript accepted for presentation at NordDesign 2024.

Contributions

The author of this thesis is the main author of Papers II, III, IV, and V. She has been the main responsible for conceptualisation, methodology, data collection, data display and analysis, writing, and reviewing of these appended papers. Her co-author in Paper II provided support in conceptualisation, data analysis, and reviewing of the paper. Her co-authors in Paper III provided support in conceptualisation, software, data analysis, and reviewing of the paper. Her co-authors in Paper IV provided support in conceptualisation, data collection, data analysis, and reviewing of the paper. Her co-authors in Paper V provided support in conceptualisation, data collection, data analysis, and reviewing of the paper.

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Chapter 1

1. Introduction

Military aircraft development is inherent to the Swedish Aeronautical industry. Saab has produced five families of fighter jet aircraft over the last eighty years. Fighter aircraft are typically multi-role platforms. The Saab Gripen aircraft is designed to be a fighter, attack, and reconnaissance aircraft. In this thesis, operational capability entails the missions the aircraft is designed to carry out. While platform capabilities entail the characteristics it is designed to have to support the various aircraft roles. Some of these characteristics include stealth, manoeuvrability and agility, and beyond visual range. Therefore, roles are facilitated by operational and platform capabilities. These capabilities in turn are supported by basic aircraft function. Typical basic aircraft functions include aerodynamics, airframe structure to sustain loads, flight mechanics and control, navigation, power management, propulsion, and thermal management. When basic aircraft function does not perform up to specification then operational and aircraft capabilities are impacted. Basic aircraft function design should commence at the early development stages. This could result in determining if a basic function can support the intended operational and aircraft capabilities early in the aircraft project. To enable this need at an aircraft developer like Saab, effective and efficient design practices are required. This thesis focuses on what effective thermal management design must entail and what practices can be implemented to realise it at Saab.

1.1. Definition of Thesis Terminology

The definition of various terms used throughout this thesis are described in this sub-section. This enables a better understanding of the rest of the thesis.

- **Vehicle systems** are the subsystems that enable an aircraft to fly safely throughout a mission. Typically, they are the systems that transfer energy to other systems, equipment, and

subassemblies in the aircraft [1]. Generally, vehicle systems that are common to most types of engine-powered aircraft include the engine, ECS, fuel system, electrical power system, actuation system, and landing and braking systems.

- An aircraft environmental control system **(ECS)** is typically designed to provide multiple functions such as air pressurization, temperature maintenance, and climate control for various systems, components, and sections of the aircraft.
- A vapor cycle system (**VCS**) is a closed-circuit refrigeration system that typically consists of an evaporator, an expansion valve, a compressor, and a condenser.
- In this thesis, a distinction is made between concept and architecture. The definition of a **concept** by Ref. [2] is adopted here. Ref. [2] defined a concept as a notional mapping between function (what the system or product does) and instruments of form (what the system or product is). On the other hand, an **architecture** is defined as a pictorial representation of the relationships between the forms and operands (the thing that the function changes).
- **Terminal heat sinks** are the destination for onboard waste heat [3]. Refs. [4, 5] noted that there are only two options for terminal heat sinks, air (ram or engine fan) and fuel.
- Thermal management **(TM)** refers to the acquisition, transport, and rejection of waste heat as considered in Ref. [5]. Thermal management system **(TMS)** refers to the suite of aircraft systems that are responsible for TM. Typically, they are the ECS, the fuel system, and any other cooling means. While **TM design** refers to the process of designing and analysing an aircraft TMS.
- **Effective TM design** entails ensuring that TMS architectures can be analysed to determine if the system can support operational and aircraft capabilities.
- A product development process (**PDP**) is a series of stages or activities exercised by a business to go from concept to commercialised product [6]. The generic life cycle model for systems or products defined by ISO/IEC/IEEE 15288:2015 in Ref. [7] consists of five main stages. They are the concept, development, production,

utilization/support, and retirement stage. In this thesis, using ISO/IEC/IEEE 15288:2015 to define the life cycle of an aircraft, **early product development** refers to the aircraft concept stage. The definition of this stage is similar to the 'concept development' stage in the generic PDP presented by Ref. [6]. At the concept development stage activities include collection of customer needs and concept feasibility investigations.

- **TM design practice** in this thesis refers to the implementation of various notions and aspects from PDP theories that can enable the use of frameworks, methods, and tools for effective TM design.

1.2. Motivation

The contemporary challenges that aircraft face include more extensive use of electrically driven technologies. This is in comparison to conventional technologies that are driven by pneumatic or hydraulic means. This leads to more electrical power (kW) demand that in turn typically leads to more cooling power (kW) demand from the TMS. There is an additional TM challenge for military aircraft with greater demand for cooling power from tactical systems [4, 8, 9, 10, and 11] with very low efficiencies in power conversion. Therefore, aircraft require greater use of their terminal heat sinks. However, a TMS of modern aircraft faces another challenge with shrinking heat sink capability. This is because of platform design aspects. Increasing use of composite materials for airframe skin over traditional metals impedes the dissipation of waste heat through skin [11-21]. Infrared and radar cross-section signatures are minimised to improve aircraft stealth by minimising the cross-sectional areas of ram air intakes [11, 12, 14, 16-19]. This reduces the amount of waste heat that can be dumped overboard through ram air. All these contemporary challenges create a strong need to commence TM design at the aircraft concept stage. This need is further exacerbated by the fact that TM can greatly impact mission operability as noted in Refs. [22,23] and it can impact aircraft performance as noted in Ref. [12]. Therefore, effective TM design must be conducted at the aircraft concept stage to determine if it

can support operational and aircraft capabilities for various platform design constraints.

Best practices must be adopted by an aircraft developer to enable effective TM design. These best practices are reflected in PDP theories. They provide methods and concepts on how companies could organise themselves and how they could collaborate with external entities in different stages of the development process. They also explore the importance of need finding and iterative and efficient design processes. On the other hand, typically literature on TM methods for architecture generation, and modelling and simulation are presented without the context of when it can be used in the PDP. Similarly, sophisticated methods for system component control optimization are presented without this context. Although, the need for TM design to be considered early in an aircraft project has been noted in Refs. [24, 25]. The use of when various methods and tools can be supportive to effective TM design is typically not presented.

The Gripen E/F are the latest versions in the latest fighter jet family at Saab, the Gripen family of fighter aircraft. The development of the first versions of the Gripen, namely A/B, commenced in the early 1980s. The general airframe configuration and vehicle systems such as the ECS and fuel system of the Gripen E/F are like its predecessors within the Gripen family.

Saab enters a new era for fighter aircraft development with its future combat air system program titled 'KFS' announced in 2024. KFS is a Swedish-language acronym that roughly translates to 'concept development of the future fighter aircraft'. The program must address the contemporary TM challenges while developing a new TMS that can support the operational and platform capabilities of the future fighter aircraft. In addition, it has been over 40 years since the development of a new fighter aircraft TMS at Saab. An industrial opportunity presents itself to develop the appropriate support to carry out TM design effectively at the aircraft concept stage.

1.3. Goals of Thesis

The primary goal of this thesis is to influence TM design practice at the aircraft concept stage.

1.4. Research Questions

Two research questions, RQ1 and RQ2 were formulated to help attain the primary goal of this thesis.
RQ1: How can effective TM design be conducted at the concept stage at an aircraft developer?
RQ2: How and when can control optimization of system components commence?

1.5. Outline of Thesis

In Chapter 1, the background and motivation of the thesis are discussed. The goals, and research questions are also defined. In Chapter 2, the research areas relevant to this thesis and areas the thesis contributes to are discussed. Chapter 3 outlines the research methodology of the thesis. Chapter 4 summarises the appended papers this thesis is based on. Chapter 5 summarises the approaches taken in the appended papers and presents the key results of the thesis. Chapter 6 provides the answers to the research questions. Finally, Chapter 7 provides the concluding remarks of the thesis.

Papers I through V form the core research work conducted in this thesis. These papers are appended at the end of this thesis.

Chapter 2

2. Frame of Reference

With the motivation, goal, and RQs for this thesis defined in Chapter 1, the areas of relevance and contribution for the thesis can be defined and described. These are the areas the thesis is developed on and onto which it contributes. An 'areas of research and contribution' diagram for this thesis is shown in Figure 1. It is created in a similar way to the one described in Ref. [26]. The main subject area in this thesis is aircraft thermal management design. Five essential areas were identified to support the main area. They are complex systems, early product development, TM methods, TM technologies, and design theories. These areas are discussed in the context of TM and vehicle systems. The various themes under these areas that are relevant to the thesis, or the thesis contributes to are also discussed.

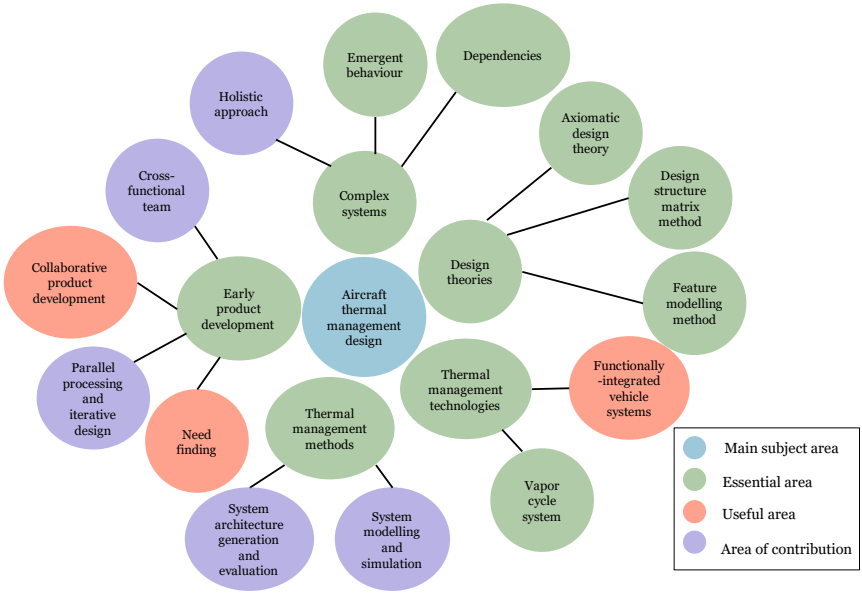


Figure 1 The areas of relevance and contribution for this thesis. The diagram is created in a similar way to the one described in Ref [26].

2.1. Complex Systems

Three main themes in complex systems literature were considered in this thesis. Literature on dependencies in complex systems and emergent behaviour were considered essential for the research conducted in this thesis. The thesis contributes to the theme 'holistic approach' under the complex systems umbrella.

2.1.1. Dependencies

Aircraft are complex systems. Depending on the aircraft type and the missions the aircraft is being designed to perform there are various factors that significantly affect vehicle system design. However, these factors may not be clearly defined at the aircraft concept stage. To identify and understand these factors complex systems theories were considered in this thesis.

The distinction between complex and complicated systems was considered in this thesis using the definitions by Ref. [2]. Apparent complexity is akin to the level of complicatedness of a system. The term 'complicated' can be used as a measure of human ability to perceive and comprehend complexity. If a system is complicated, then it has high apparent complexity, and complicated systems are hard for humans to comprehend. Using this notion, an aircraft vehicle system designer would find it far less complicated to analyse the architecture of a vehicle system as a standalone system than to analyse it when it is integrated in the aircraft. However, the former is rarely ever the case. To understand integration and 'interconnectedness' within a complex system such as an aircraft, the concepts, 'complex interactions' and 'tight couplings', coined by Ref. [27] are applicable. These two concepts were considered in the context of aircraft vehicle systems.

Complex interactions and tight couplings are inherent to aircraft, and they play a significant role in the design choices for aircraft vehicle systems. Regarding the concept of complex interactions, aircraft consist of many interconnected subsystems that interact with each other to perform various functions. This leads to many connections between components within a subsystem as well as connections between different subsystems.

Further, the internal configuration of the aircraft may also result in complex interactions between systems and equipment that are physically placed near each other in the aircraft even if they may not be physically connected to one another.
Regarding the concept of tight couplings, there are many tight couplings between a vehicle system and the system(s) it depends on or between the vehicle system and the subsystems that depend on it. If a vehicle system such as the ECS depends on the engine for bleed air supply, then what are the redundancies in place if the bleed air supply system fails? If a subsystem depends on the electrical power system, then what are the redundancies in place if the power system fails? Therefore, there is a strong need to understand dependencies within a complex system in the early phases of development. This is to avoid the unanticipated and unwanted emergent behaviour.

2.1.2. Emergent Behaviour and Holistic Approaches

Ref. [28] noted that system of systems possesses five characteristics that differentiate them from conventional complex systems. One of these characteristics is emergent behaviour. Emergent properties of vehicle systems may be impacted when they are integrated into an aircraft. They may be impacted when the aircraft operates in various missions in different environmental conditions. Therefore, a holistic approach to vehicle system development is needed. One holistic approach to product development in the context of system of systems was proposed by Ref. [29]. The approach spanned from end-user needs to the subsystem design space. This thesis considers how customer needs impact vehicle system design and sizing (in terms of weight, volume, and power output).
Ref. [30] considered several aspects that challenge engineering system design, where an engineering system is considered being complex in technical and human terms. Some of these aspects have been considered in vehicle system design and evaluation in this thesis. One aspect considered is the system dimension in terms of decomposing an aircraft into its subsystems that are further decomposed into its constituent components. On the other end of the scale the aircraft is a

constituent of the 'product' that is offered to the customer in terms upgradability and support through the complete lifecycle of the aircraft. Another aspect considered was nature of innovation and its impact on vehicle system evaluation. Ref. [30] scaled innovation from evolutionary to disruptive. They noted that disruptive aerospace technology can make it difficult to compare design solutions. It would only be possible if system-level performance, behaviour, and functionality are compared on identical bases. Disruptive technologies for TMS are discussed in further detail in Section 2.3. This thesis considers vehicle system evaluation and TM design from a holistic perspective.

2.2. Early Product Development

Four main themes in PDP literature were considered in this thesis to understand best practices in early product development. Literature on 'collaborative product development' and 'need finding' were considered useful for this thesis. The thesis contributes to the themes 'cross-functional team' and 'parallel processing and iterative design'.

2.2.1. Cross-functional Teams

The need for cross-functional teams may vary during a project [31]. It was noted that they may be more useful in the early phases of the project when setting product specifications. A study case at Honda showed that cross-fertilized teams resulted in new ideas and concepts [32]. Cross-functional teams or integrated product teams (IPT) are essential to early aircraft development. This is due to the interconnectedness and large number of interdependencies in a complex system such as an aircraft as noted above. There can also be several dependencies for information needed for design. These dependencies can be better understood in cross-functional teams. This thesis addresses cross-functional team formation through industry-based data.

2.2.2. Collaborative Product Development

Ref. [33] noted that companies achieved significant improvements in projects when suppliers were involved compared to projects in which they were not involved. Improved collaboration between aircraft developers and their suppliers and customers is a significant enabler for contemporary aircraft development [34]. Activities such as design, operation, and maintenance of contemporary and innovative aircraft need efficient collaboration among multidisciplinary engineers and other entities [35, 36]. Communication and collaboration at the concept stage between vehicle system designers at the aircraft developer and vendors are briefly addressed in this thesis.

2.2.3. Parallel Processing and Iterative Design

Traditionally product development has proceeded in a sequential manner in a succession of phases for a long period of time [31]. Parallel processing is demonstrated as an important feature of the stage-gate model where activities take place concurrently with different functions from across the firm [37]. Parallel incremental development is inherent to the development of complex products such as aircraft. At the onset of the Cold War, the complexity and ambition of systems were increasing in the US military. This weakened the traditional single-discipline, linear approach to system development that entailed making one part of the system, then the next, and so on. In turn, this gave rise to the systems engineering-integration approach that provided a systematic, multi-disciplinary approach to systems development which was concurrent and not sequential [38]. One approach to developing complex systems is the integration-driven development approach described in [39]. In integration-driven development, interfaces are very important, and they should be established as early as possible for large projects, with several teams working in parallel. An iterative approach to product development is proposed in the stage-gate model in Refs. [37, 40]. The third principle of design by Ref. [41] states that the gradual and consistent evolution in product

development is a result of iterations – repetition of activities to obtain the desired result.
The iterative process is essential to aircraft development. The iterative process is tied into milestones such evaluation reviews conducted in the concept and development phases. The reviews can result in multiple iterations of the platform itself (aircraft-level) or its subsystems. This thesis addresses iterative design and parallel processing for vehicle systems and in a greater context for TMS that could be carried out at the aircraft concept stage.

2.2.4. Need Finding

Ref. [42] noted that the initial step in complex system design is to determine the customer attributes or needs. Ref. [43] described how being need-oriented rather than being solution-oriented keeps all possible solutions open for consideration. Ref. [44] noted that development project managers are usually concerned with problem identification since solving them becomes more costly and time-consuming as the project progresses through its phases. A thorough process for need finding is required early in aircraft development. This is because aircraft development is affected by long lead times [29, 30, and 45] and forecasts and predictions become more uncertain the further the development prolongs. The information required from the customer at the early development stage in the context of TM is addressed in this thesis.

2.3. Thermal Management Methods and Technologies

Four main themes were considered under the two areas of TM methods and TM technologies. Under the area of TM methods, this thesis contributes to both themes, system architecture generation and evaluation, and system modelling and simulation. On the other hand, under the area of TM technologies, understanding VCS technologies and their use in aircraft was considered essential to this thesis. And understanding functionally integrated vehicle systems was considered useful.

In recent years, several methods for TMS architecture generation [25, 46-48] and TMS modelling and simulation [10, 12-16, 19, 24, 49-51] have been presented. However, aircraft are complex systems with several interdependencies that are typically not defined at the concept stage. Obtaining the information to populate these methods may require several different steps at an aircraft developer like Saab. Addressing this industrial challenge is the one of the main research aspects of this thesis.

Nowadays, traditional, federated vehicle systems are being replaced by more-functionally integrated vehicle systems as noted in Refs. [52 and 53]. A more-functionally integrated vehicle system carries out functions that would usually be carried out by multiple traditional, federated vehicle systems. Emergent properties are used to evaluate a vehicle system concept. For an aircraft developer, predicting the emergent properties of a more-functionally integrated vehicle system might prove more challenging than predicting those of traditional, federated vehicle systems. This is because tacit knowledge and historical data for traditional, federated vehicle systems may be available at the aircraft developer. However, with the unavailability of data and knowledge for more-functionally integrated vehicle systems, the current evaluation approach may need to be improved. This need for improvement is addressed in this thesis from a holistic perspective.

Over the last three decades, VCS have emerged as a more efficient alternative to conventional aircraft cooling systems such as air cycle machines (ACM). In comparison to ACM, VCS are more energy efficient. They do not require a source of high-pressure air from engine bleed air or conventional ram air for operation [54-57]. They have successfully been used on operational aircraft. Successful applications of VCS on military aircraft include the Lockheed Martin F-22 [55, 58-61]. Civilian aircraft applications include Embraer Phenom 100 and Embraer Phenom 300 [62]. In recent years, several studies on VCS for aircraft applications have focused on analysing various control strategies for the system. These include either modelling and simulation or experimental set-up studies [57, 63-67]. However, understanding when and how system component control optimization can

commence is essential in aircraft development and it is addressed in this thesis.

2.4. Design Theories

Ref. [68] noted a lack of methodological support for architecture generation at the early product development phase. While Ref. [69] noted a comprehension of engineering design activities is enabled by design theories. Aspects of three theories, axiomatic design theory, DSM, and feature modelling were utilized for architecture generation and analysis in this thesis. Therefore, all these theories were considered essential to the thesis.

2.4.1. Axiomatic Design Theory

According to Ref. [70] the design world consists of four domains, namely the customer domain, the functional domain, the physical domain, and the process domain. The customer and functional domains characterise what is to be achieved while the physical and process domains characterise the design solution. The design solution is how the requirements specified in the customer and functional domains are satisfied.

Customer attributes (CAs) represent what the customer is seeking in a product or a system and are represented in the customer domain. The customer needs are represented in terms of functional requirements (FRs) in the functional domain. Design parameters (DPs) are conceived in the physical domain to satisfy the FRs. The DPs refer to the key physical variables of a system or product. Finally, in the process domain a process is created, and it is defined by process variables (PVs) to produce the product or system defined by DPs. In this thesis, PVs were not considered. Further, the terms customer domain and customer attributes were changed to stakeholder domain and stakeholder attributes, respectively. This is because vehicle system architecture generation considered the aircraft designer as a ‘stakeholder’ of the vehicle system. The aircraft designer provides attributes for the vehicle system.

To obtain acceptable design solutions when using axiomatic design theory, the independence axiom and the information axiom must be satisfied [70]. The independence axiom is satisfied when each FR for a product or a system can be satisfied without impacting any of the other FRs. The information axiom is employed when there are many acceptable designs that satisfy the independence axiom. The second axiom can be used to select the best among those designs. The information axiom states that the best design is the one with the least information content. Therefore, the information axiom provides the theoretical foundation for design optimisation and robust design.

2.4.2. Design Structure Matrix Method

A three-step method to increasing understanding of complex systems is proposed by Ref [71]. First the complex system must be decomposed into its subsystems about which more is relatively known. Then the relationships between the subsystems that give rise to the behaviour of the system must be noted. Finally, the inputs to the system and the outputs from the system and their impact on the system must be noted. These steps are captured in the DSM method. In this thesis, the DSM method was used to analyse dependencies between system component and between TM design activities.

2.4.3. Feature Modelling

A feature is defined as a distinct user-visible aspect, quality or characteristic of a system [72]. Feature modelling is a domain analysis technique typically used to define software product-lines and system families [73]. In this thesis, feature modelling was used to create a feature model for various aspects of an aircraft ECS.

Chapter 3

3. Research Methodology

This chapter describes the research methodology of this thesis. A general description of research methodology is provided followed by the methodology specific to the thesis.

According to Ref. [74], a method is simply a systematic approach. They define a research method as an established approach to attain the maximum knowledge possible about something that is not known. And they define a methodology as a study of methods that connects assumptions made about the world and how it may be studied. Therefore, a methodology is the overarching logic for how to perform a study.

Six types of research methods and data collection methods were identified by Ref. [74]. These are all shown in Figure 2. Two types of data and three types of data analysis methods identified by Ref [74] are also shown in Figure 2. In this thesis, experiment, survey, case study, and design research methods were used in Papers I through V. Both qualitative and quantitative data was collected in this thesis. Data collection methods included interviews, questionnaires, workshops, and document studies. Finally, qualitative data analysis was used to analyse the qualitative data collected in the thesis. Qualitative data analysis consists of three main steps that are carried out iteratively and cyclically along with data collection. They are data display, data reduction, and verifying and/or drawing of conclusions [75].

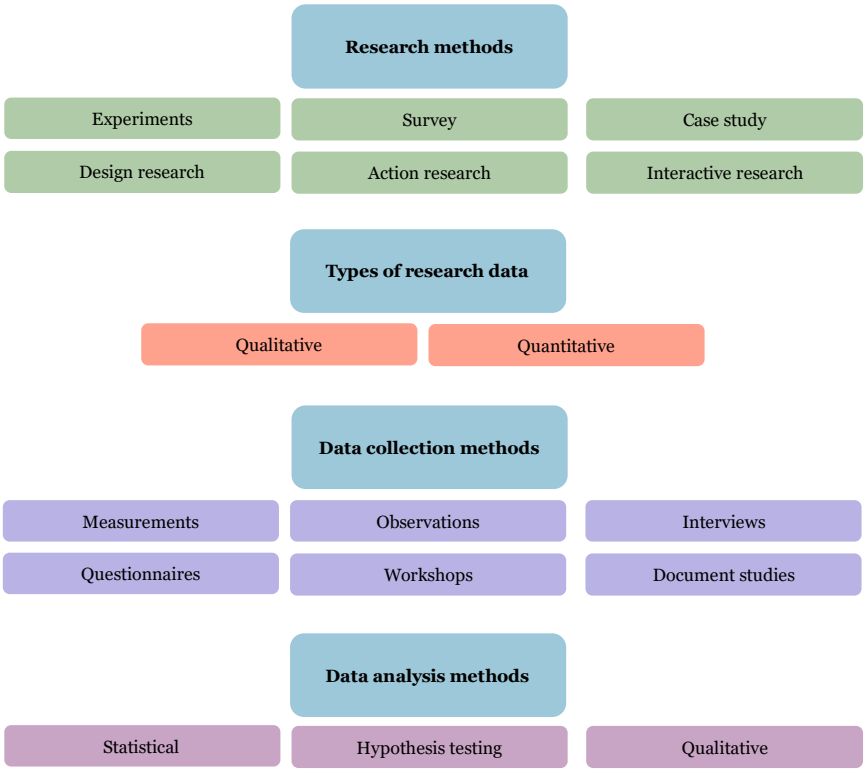


Figure 2 Types of research methods, research data, data collection methods, and data analysis methods. The classification was created based on Ref. [74].

The type of research question defined for the study drives the type of research method that is chosen [74]. Experiments are typically used to find cause-effect relationships to determine for example which variables have the largest impact on the outcome of the process or system. It could also be used to analyse an event or artifact in relation to one or more variables [74]. Surveys are suitable for carrying out broad studies where several aspects are to be examined in relation to a few variables. Surveys are used when the goal is to be descriptive or explanatory. Ref. [74] defines them as non-experimental, sample-like, large-scale studies. They also note that they differ from questionnaires because a survey can employ multiple data collection methods. Case studies are exploratory in nature and are well-suited when more comprehensive knowledge is required or to be described for a situation, event, or phenomenon. They are typically used to answer how

and why questions. Design research entails the activities needed for realising a product from a need, idea, or technology that fulfils the needs of its users and other stakeholders [26]. Ref. [26] states that design research has two aims. One is to generate knowledge about the product development process. The second is to create support that enables and improves product development and the consequent results.

3.1. Design Research Methodology

Design research methodology (DRM) was created by Ref. [26] and it is a comprehensive research approach consisting of several research methods and techniques for data collection [74]. The goal of DRM is to generate understanding and support to improve design research that in turn pursues to understand, support, and develop product and process development [26]. The DRM framework is an iterative process with four main steps as shown in Figure 3. Each step has a means to fulfil it and an outcome from it.

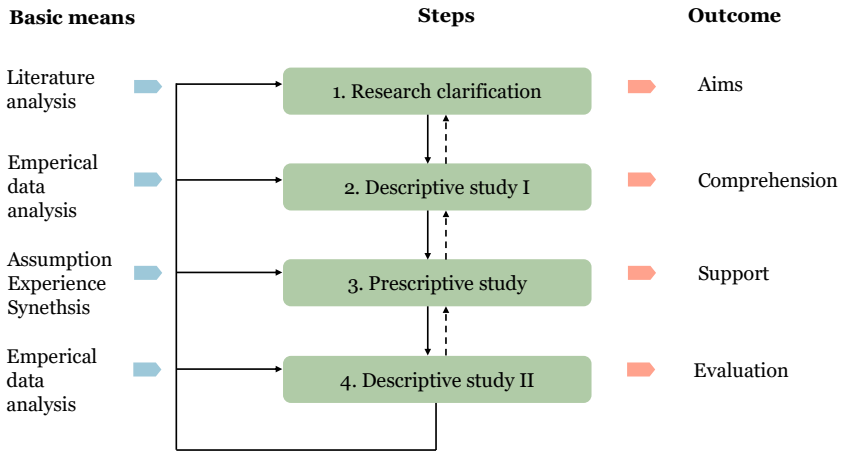


Figure 3 Design research methodology framework by Ref. [26].

Research clarification (step 1) entails identifying the areas of relevance and contribution. It also entails defining the research goals, scope, plan, and defining the evaluation criteria for the research outcomes. A 'review-based descriptive study I' (step 2) entails an in-depth literature review in the areas related to the research goals

defined in step 1. A 'comprehensive descriptive study I' (step 2) is conducted when a review-based one indicates a lack of comprehension about the research topic of interest. Then, further empirical studies are needed to improve the understanding. A prescriptive study (step 3) entails creating design support by using the in-depth understanding gained in step 2. The support reflects the initial desired research goals defined in step 1. Finally, a 'descriptive study II' (step 4) entails empirical studies to evaluate the degree to which the design support created in step 3 contributes to the goals defined in step 1. The four steps can be combined in seven ways to create seven types of design research projects. The first three types of projects are shown in Figure 4.

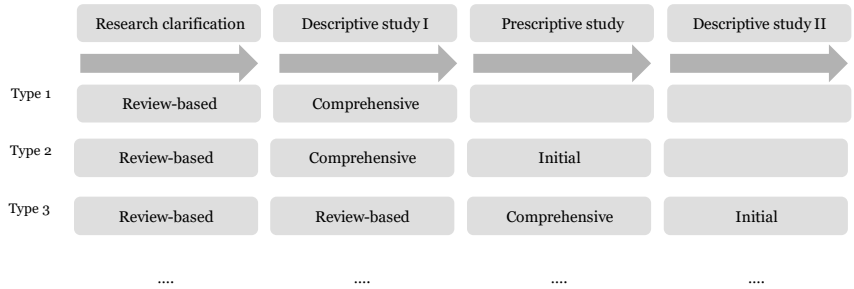


Figure 4 Three of the seven different types of design research projects that are presented in Ref. [26].

3.2. Methodology Framework for the Thesis

The two research questions formulated in this thesis and stated in Chapter 1 drove the need for a type 2 design research project for this thesis. The methodology framework for the thesis is shown in Figure 5 . The review-based research clarification stage of the thesis, the comprehensive descriptive study I, and the initial prescriptive study that resulted from the thesis are described below.

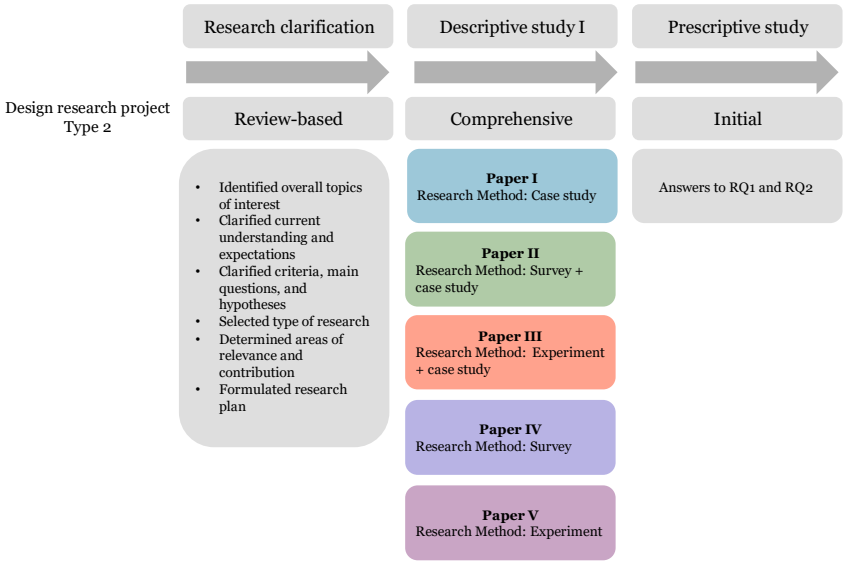


Figure 5 The research methodology framework for the thesis.

Review-based research clarification was carried out:

- To identify the overall topics of interest, i.e., aircraft TM needs for future programs at Saab.
- The current understanding and expectations were obtained from literature reviews conducted on various contemporary topics related to TM.
- Goals and research questions were defined based on the work carried out that is described in the first two points.
- The type of research was chosen based on the research questions.
- The areas of relevance and contribution were determined, and they formed the frame of reference for the thesis as described in Chapter 2.
- Finally, the overall research plan for the thesis was created. This led to the creation of the individual plans for each of the five appended papers.

The five research studies presented in the appended papers contributed to the **comprehensive descriptive study I** of the thesis. The chosen research method based on the specific RQ(s), type of data,

data collection method, and data analysis method for each of the five appended papers is summarised below.

- Paper I
    - RQ specific to paper: How can feasible vehicle system design be conducted at the aircraft concept stage?
    - Research method: Case study
    - Reason for chosen method: For exploration on how to design at the concept stage.
    - Type of data: Qualitative
    - Data collection method: Literature review + retrospective data collection through unstructured interviews at Saab
    - Data analysis method: Qualitative data analysis

- Paper II
    - RQ specific to paper: How can vehicle system evaluation can be conducted at the aircraft concept stage in a holistic manner accounting for multiple aspects?
    - Research method: Survey + case study
    - Reason for chosen methods: To describe and explore a contemporary event considering evaluation of new technologies such as more-functionally integrated vehicle systems.
    - Type of data: Qualitative
    - Data collection method: Semi-structured interviews + document studies
    - Data analysis method: Qualitative data analysis

- Paper III
    - RQs specific to paper: How do operating conditions, requirements, and constraints impact the availability of aircraft cooling system parameters? How can a parameter tuning study be set up? How can the detailed performance of a cooling system consisting of a VCS be set up using computational modelling and simulation?

How can the simulation results be used to optimize the control strategy of system components?
    - Research method: Experiment + case study
    - Reason for chosen methods: To understand the cause-effect relationship between requirements and design variables. To analyse relationships between computational model set-ups and results. To explore modelling and simulation for a VCS.
    - Type of data: Quantitative
    - Data collection method: Computational modelling and simulation
    - Data analysis method: -

- Paper IV
    - RQ specific to paper: What does effective TM design at the aircraft concept stage entail?
    - Research method: Survey
    - Reason for chosen method: To understand and explore what effective TM design entails at the concept stage at an aircraft developer.
    - Type of data: Quantitative + qualitative
    - Data collection method: Workshop (including semi-structured interviews).
    - Data analysis method: Qualitative data analysis

- Paper V
    - RQ specific to paper: How can the usefulness of AI support for aircraft ECS design be evaluated?
    - Research method: Experiment
    - Reason for chosen method: To determine the cause-effect relationship for a contemporary event which was to evaluate the usefulness of an AI tool in supporting ECS design.
    - Type of data: Qualitative
    - Data collection method: Workshop + questionnaire (in the form of a Turing test)

- Data analysis method: Qualitative data analysis

Finally, the answers to the two RQs of the thesis are provided in Chapter 5. An **initial prescriptive framework** for effective TM design at the aircraft concept stage is presented and described. A summary of which papers answer the two RQs of the thesis is presented in Table 1.

Table 1 A summary of which appended papers answer the two thesis research questions.

Paper	RQ1	RQ2
I	x	
II	x	
III	x	x
IV	x	x
V	x	

Chapter 4

4. Appended Paper Summarisations

The results and findings of the five appended papers are presented in this chapter.

4.1. Paper I: Harnessing Vehicle System Design Complexity

4.1.1. Objective

The objective of this study was to provide design support to vehicle system designers at the aircraft concept stage. Support that can account for the significant factors that affect vehicle system design but may not be clearly defined at the concept stage. Support that can enable feasible vehicle system architecture generation.

4.1.2. Approach

Firstly, various complex systems theories were considered in this paper to understand the significant factors. Aspects from complex products and systems (CoPS) theory by Ref. [38] were used to understand the role of an aircraft vehicle system designer at an aircraft original equipment manufacturer that designs and builds aircraft. Aspects of normal accident theory by Ref. [27] and system architecture theory by Ref. [2] were used to identify the factors that significantly influence the design choices for a vehicle system architecture. Secondly, aspects from axiomatic design theory, feature modelling method, DSM method, and the systems thinking approach were used to create a framework in this paper. The framework was based on industrial experiences at Saab, and it used a manned fighter aircraft ECS as a case study.

4.1.3. Results and Findings

The framework created in this study consisted of eight steps as shown in Figure 6 below. It can be used recursively to generate feasible vehicle system architectures while accounting for the significant factors that affect vehicle system design but might change during the concept stage.

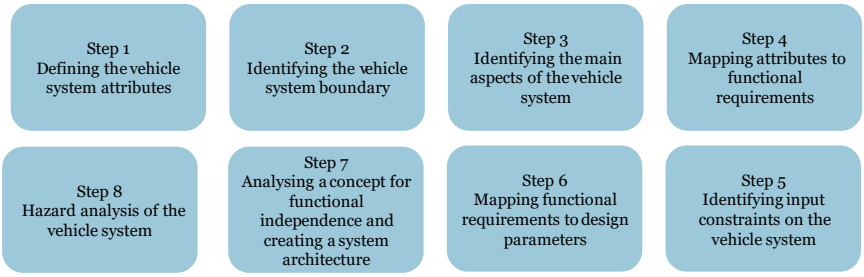


Figure 6 The eight-step framework for feasible vehicle system architecture generation at the aircraft concept stage.

4.2. Paper II: Holistic Evaluation Approach for Vehicle Systems

4.2.1. Objective

The objective of this study followed on from that of Paper I to support vehicle system designers at the aircraft concept stage in system evaluation. The objective was to understand what aspects of an aircraft project impact the emergent properties of vehicle systems and thereby make holistic predictions about these aspects.

4.2.2. Approach

A qualitative study on the Saab Gripen E/F aircraft project was conducted using retrospective data collection methods. The qualitative study was used to understand how the evaluation of a vehicle system is carried out at an aircraft developer. A manned fighter aircraft ECS was the system of interest in this study. Data was collected from formal product and project documentation and semi-structured

interviews with six engineers. Data collected through documentation was limited to system specification documents of the ECS of Gripen C/D and Gripen E/F. All six engineers interviewed worked in the Gripen E/F project. They included an operational analysis engineer, two aircraft concept design engineers, two ECS-expert engineers, and one design and installation engineer. The data was merged and synthesised to describe how aspects of design in an aircraft project impact the emergent properties of vehicle systems.

4.2.3. Results and Findings

The collected data demonstrated that the emergent properties that a vehicle system are evaluated for are impacted by various levels of design in an aircraft project. Four design levels were considered in this paper, namely, operational, aircraft, subsystem, and component. The impact of these levels on the emergent function, emergent performance, and emergent safety of a manned fighter aircraft ECS is summarised in Table 2. These three criteria were chosen because unlike some of the other evaluation 'ilities', the collected data indicated that these three criteria were heavily impacted by aspects from the design levels considered in this paper.

Table 2 Aspects from the operational, aircraft, subsystem, and component design level of an aircraft project that impact the emergent function, emergent performance, and emergent safety of the ECS case study in this paper.

		Aspects from Design Levels in an Aircraft Project			
		Operational	Aircraft	Subsystem	Component
Emergent Properties of the ECS case study in this paper	Emergent Function of the ECS	• Operation in hot, humid climate	• Aircraft modes • Aircraft Functions	• ECS providers • ECS consumers	• ECS components
	Emergent Performance of the ECS	• Altitude and speed envelope for aircraft • Missions flown by aircraft • Frequency of each mission	• Aircraft modes	• Surrounding equipment and/or systems	• Operating range of each ECS component
	Emergent Safety of the ECS	• Maneuvers performed in a mission	• Aircraft modes	• Operating ranges of ECS consumers • Monitoring and control of air to consumers Design and installation of airframe	• Probability of failure of each ECS component • Redundancies for monitoring and control of air

4.3. Paper III: Parameter Tuning Study

4.3.1. Objective

This was a multi-objective study. The first goal of this paper was to demonstrate how operating conditions and requirements impact the availability of aircraft cooling system parameters. The second was to demonstrate how to set up a parameter tuning study. The third to demonstrate a computational modelling and simulation strategy for an aircraft cooling system consisting of a VCS. Thus use this strategy to not only find the limits of the system for various operating conditions but to also obtain a detailed performance analysis of the system components. The final goal was to demonstrate how simulation results can be used to optimize the control strategy of the system components.

4.3.2. Approach

To meet these goals a two-part method was used in this paper. A cooling system for a military surveillance aircraft was used as a case study.

The first part of the method was a parameter tuning study set-up that preceded the modelling and solving strategy for the cooling system. The workflow for the tuning study set-up is pictorially shown in Figure 7.

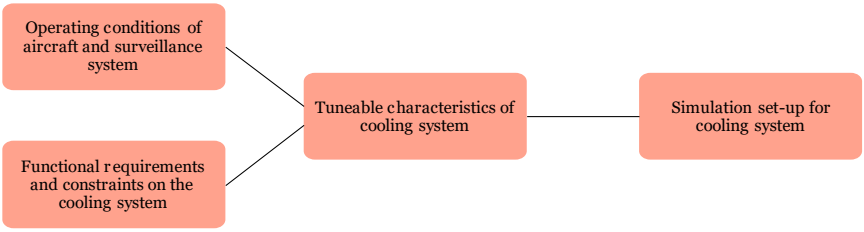


Figure 7 Method workflow for parameter tuning study set-up for the cooling system.

First, the operating conditions of the aircraft and surveillance system were identified and quantified. They were based on the detection and identification characteristics of the surveillance system. Two operating points were defined in terms of altitude and Mach number. Similarly, the requirements on the cooling system were identified. The main requirement on the system was to provide cooling power to the surveillance system at both operating points for a range of surveillance heat flow rates from 10kW to 70 kW. The operating conditions and requirements were used to identify the tuneable characteristics of the cooling system. Finally, this led to the cooling system simulation set-up shown in Figure 8. Compressor speed (Hz) drove the simulation set-up for this study. Surveillance heat flow rates at operating points 1 and 2 were tuned from 10 kW to 70 kW in increments of 10 kW. At each heat flow rate, the compressor speed (Hz) was tuned from 'compressor speed 1' to 'compressor speed n' to investigate if the main requirement was met. At each compressor speed from 1 to n, a dynamic simulation was run until the solution reached steady state. The dynamic simulation set-up was done in the second part of the method.



Figure 8 The set-up of the simulation study of the cooling system.

The second part of the method for this study entailed the modelling and solving strategy for the cooling system. The cooling system was modelled and simulated in Modelon Impact. Modelon Impact is a systems design environment that supports system-level modelling, simulation, optimization, and analysis with Modelica-language based model libraires.

4.3.3. Results and Findings

The parameter tuning method resulted in the maximum cooling capacity of the system being determined at both operating points. Along with the overall performance of the system, the performance of components was also obtained. By modelling the complete system in this study and choosing the appropriate level of detail for the system components, a more realistic limit of the cooling capacity was obtained. Finally, this paper demonstrated that for each surveillance heat flow rate by running the compressor at the lowest possible speed while not violating the main requirement reduced the power output (and thereby heat flow rate) from the compressor. Therefore, the results from this two-part method can be used to design a control strategy for the compressor of the cooling system. In a larger context, the two-part method and the results analysis can serve as a preliminary method for aircraft VCS control optimization studies.

4.4. Paper IV: Cool Planes for Hot Missions

4.4.1. Objective

The main objective of this study was to demonstrate what is required for effective TM design at the aircraft concept stage. The second objective was to create support to enable effective iterative TM design at the aircraft concept stage. Thus, be able to help an aircraft developer determine if a TMS can support the intended operational and aircraft capabilities at the concept stage.

4.4.2. Approach

To fulfil these objectives, a workshop was conducted at Saab with a cross-functional team for a pre-defined scenario for TM design. The workshop was an efficient method to fulfil the goals of this study. It was chosen over other data collection techniques like individual or group interviews with experts from various functional groups. A workshop allowed the cross-functional team to work together to determine the order of the information and the various functional groups needed to provide the information for TM design. The workshop had three main phases, the set-up phase, the execution phase, and the post-workshop phase. In the workshop set-up phase, first the workshop goals were defined to align with the goals of the study. Then data was collected to define the workshop scenario and finally the workshop cross-functional team was defined. The goals of the workshop were as follows:

- To create a common timeline for the steps needed to be carried out for TM design at the aircraft concept stage.
- To understand the dependencies between the various steps of the TM timeline.
- To determine the various functional groups that need to be involved in each step of the TM timeline.

Aircraft capabilities and operational capabilities were defined in the scenario. An ECS and fuel system architecture were also defined and together with other cooling means formed the TMS for the study. The operational capabilities included a mission profile that included the

worst aircraft operating conditions from a TM perspective. The conditions were defined in terms of Mach number, altitude (m), fuel mass flow rate (kg/s), and time (in minutes) spent in the worst conditions. Further, cooling power demand (kW) from flight-critical and mission-critical systems that was to be met throughout the mission segments with the worst conditions was also defined.

The cross-functional workshop team consisted of seven experts from the various aircraft project design levels defined in Table 2. The definition of the workshop roles and design levels each expert represented was based on data collected for the workshop scenario. An additional level was added to this study above the operational level called the product level. The product level represents the aircraft as a product to the customer. The product includes the complete package from customer requirements to development into operational, maintenance, and upgradability support of the aircraft. Each of the seven experts played a different role in the workshop and belonged to one of the five levels. This is shown in Figure 9. The liaising expert represented the product level in the workshop. He represented the liaising role between the aircraft developer and the customer. The operational analysis expert and the overall aircraft design expert represented the operational and aircraft level, respectively. The subsystem level was represented by the ECS, fuel system, and sensor suite expert. The sensor suite expert represented all mission-critical and tactical systems of the aircraft. Finally, the component level was represented by the heat exchanger expert.

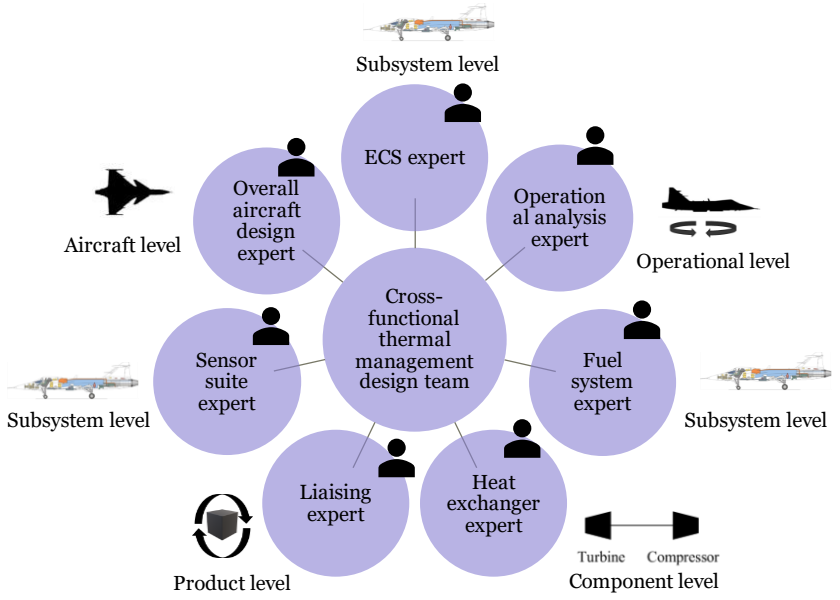


Figure 9 The roles and design level representation of the seven experts of the cross-functional thermal management design team.

In the workshop execution phase, each of the workshop goals were accomplished through a series of three workshop sessions. Following the completion of the sessions, in the post-workshop phase the results from all three sessions were then analysed.

4.4.3. Results and Findings

The workshop resulted in a framework that can be used to conduct effective TM design at the concept stage at an aircraft developer like Saab. The framework presents a detailed understanding of the steps to be carried out iteratively for TM design by a cross-functional team. It also provides the dependencies for these steps and the various functional groups that need to be involved in each step. All the functional groups external to the core team that need to be involved in TM design at Saab are shown in Figure 10. The results showed that customers and vendors are also part of TM design at the concept stage. The iterative process can be used at the aircraft concept stage to understand TM design implications on aircraft and operational capabilities. The framework can also complement other methods presented in literature for TMS architecture generation, and modelling and simulation

of aircraft TM. Finally, the workshop methodology can also be used to obtain similar frameworks for design of other basic aircraft functions at the concept stage.

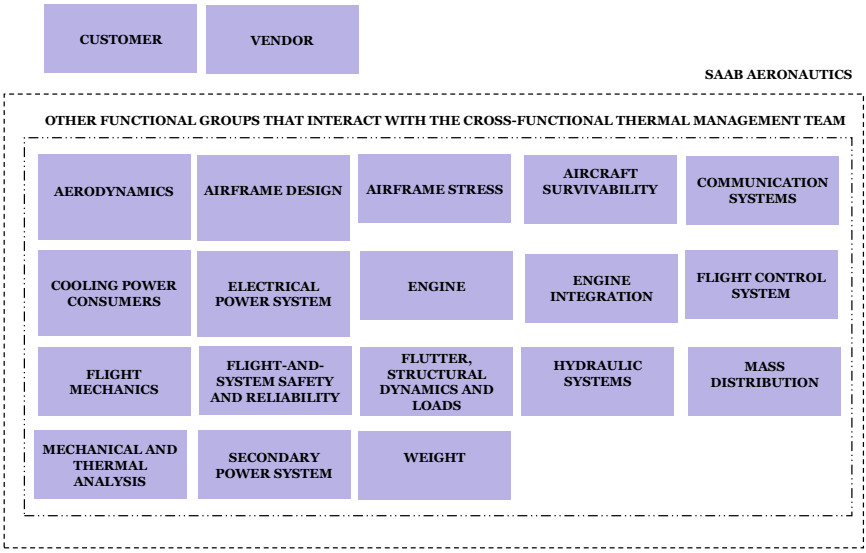


Figure 10 All the functional groups external to the core thermal management (TM) team that need to be involved in TM design at Saab. Customers and vendors are also involved, and they are external to the Saab Aeronautics organization.

4.5. Paper V: Chilling with ChatGPT

4.5.1. Objective

The objective of this study was to demonstrate how to evaluate and analyse the usefulness of ChatGPT in manned fighter aircraft ECS design at Saab.

4.5.2. Approach

To meet this objective a method was designed for this study that included a Turing test that was conducted at the Aircraft Vehicle Systems department at Saab. Four entities (three humans and one AI-tool) were part of the data collection in this study. They included three engineers and ChatGPT NEO. ChatGPT NEO is a variant of the GPT

model developed by OpenAI where NEO stands for nearly efficient optimization. It is specifically designed to be more resource-efficient and environmentally friendly compared to larger models like GPT-3, while still offering strong performance in natural language understanding and generation tasks. The three engineers chosen for this study work with early design of fighter aircraft ECS at Saab. Engineer 1 was employed at Saab for 29 years of which 25 years were spent working with concept design, detailed development, and operational support for aircraft vehicle systems. Engineer 2 and Engineer 3 were employed for 28 years and 7 years, respectively doing similar tasks to Engineer 1 in their roles at Saab. All three engineers worked on the same project at Saab at the time the data was collected.

A Turing test was conducted on the four participants of the study to collect data. First, ChatGPT and engineer 1 were each separately tasked with generating an ECS architecture using some steps from the framework in Paper I. They were also tasked with listing evaluation criteria and design variables for the architecture they each generated. Then engineers 2 and 3 conducted a blind evaluation of the responses from ChatGPT and engineer 1 for some tasks. They were provided with a relevance ranking for evaluating the responses. All three engineers were familiar with the framework in Paper I. However, ChatGPT was not explicitly informed about Paper I. The paper was not uploaded to ChatGPT during its official task session with the first author.

There were ten tasks performed by ChatGPT and engineer 1. Framework steps 1, 3, 4, 5, 6, 7, and 8 in Paper I are similar to tasks 1 through 8 in this study and are described in Figure 11. The first six tasks focus on architecture generation. The last four tasks of the study focus on architecture analysis and evaluation.

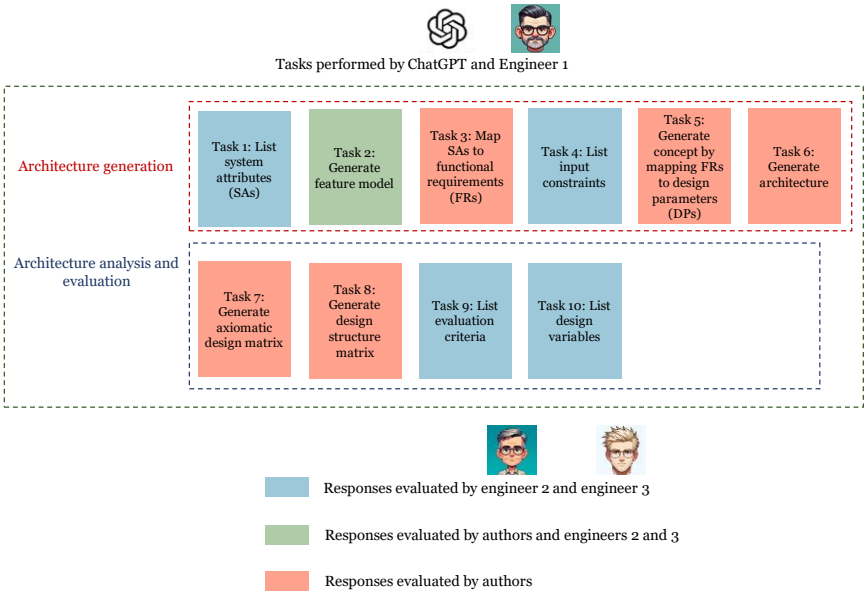


Figure 11 The tasks performed by ChatGPT and engineer 1. The responses to these tasks were evaluated by engineers 2 and 3, and the authors of this paper.

4.5.3. Results and Findings

The architectures generated from the concepts by ChatGPT and engineer 1 are shown in Figure 12. Compared to the generic architecture by ChatGPT, that by engineer 1 seemed to be influenced by his experiences working in other military aircraft projects at Saab as it contains specific components with specific connections. While the architecture by ChatGPT can be used as a generic template to test combinations of different components.

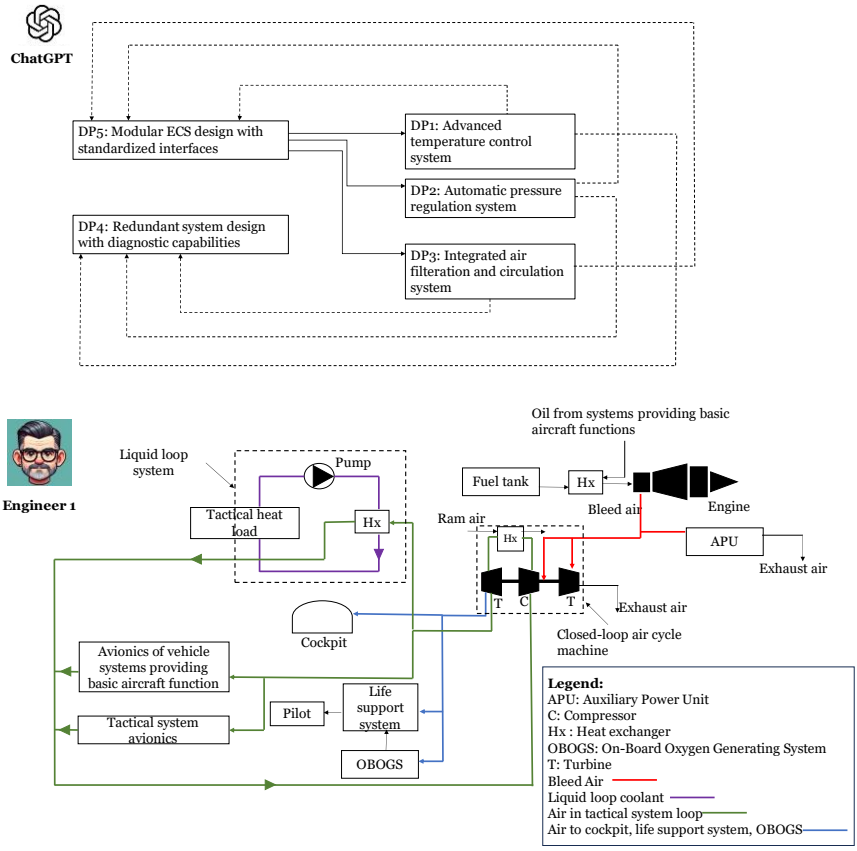


Figure 12 Architectures of a manned fighter aircraft environmental control system generated by ChatGPT (top) and engineer 1 (bottom).

Engineer 2 tended to rank the responses from engineer 1 with greater relevance than those from ChatGPT. On the other hand, engineer 3 mostly gave all responses the same ranking noting that they all had equal relevance. It could be deduced that ChatGPT was able to deceive engineer 3 but not engineer 2. Therefore, ChatGPT passed the Turing test with engineer 3 but not quite with engineer 2. This could be due to the large discrepancy in work experience at Saab between engineers 2 and 3 with the former having four times more experience than the latter. Therefore, specific system attributes, constraints, evaluation criteria, and design variables inherent to aircraft vehicle systems at Saab have not been instilled in engineer 3 yet. Therefore, engineers 1 and 2 have gained a lot more tacit design knowledge at Saab than

engineer 3. To train engineer 3 in general vehicle system design and in design aspects unique to Saab, an AI tool like ChatGPT could prove useful. It can be trained with company-specific design aspects. Therefore, engineer 3 could catch up to the Saab-level of knowledge of engineers 1 and 2 faster. Further, ChatGPT or another AI tool could assist engineers 1, 2, and 3 in designing a manned fighter aircraft ECS. The method designed for this study could be applied to evaluate the usefulness of ChatGPT or other AI tools for design of other aircraft subsystems. The results of this study and future studies like it could be used by companies like Saab to determine what knowledge AI tools need to be trained on to pass on company-specific expertise to future generations.

5. Summary of Approaches and Key Results

In this chapter the approaches taken in the appended papers are briefly described and the key results of the thesis are presented.

5.1. Summary of Approaches

Using the aircraft project design levels defined in Paper II and Paper IV, the approach (top-down, bottom-up, or both) taken in each appended paper is summarised. This is shown pictorially in Figure 13.

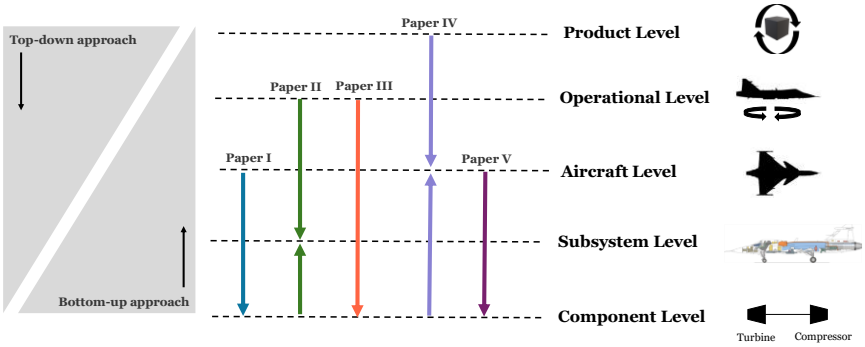


Figure 13 Five design levels of an aircraft project identified in this thesis. They are product, operational, aircraft, subsystem, and component. The five appended papers in this thesis use these levels to either take a top-down, bottom-up, or both approaches to address the challenges with thermal management design at the aircraft concept stage.

In Paper I, a top-down approach was taken to define the attributes of a vehicle system such as an ECS from an aircraft level. This was the first step of the eight-step framework for vehicle system architecture generation shown in Figure 6. In Paper II, a top-down approach from the operational level to the subsystem level and a bottom-up approach from the component level to the subsystem level were taken simultaneously. This was done to understand how emergent properties that an ECS are evaluated for are impacted by these design levels. In Paper III, a top-down

approach was taken to define the operational conditions based on the detection and identification characteristics of the aircraft surveillance system. This was part of the parameter tuning study set-up that led to modelling and simulation of the cooling system. In Paper IV, a top-down approach from the product level to the aircraft level and a bottom-up approach from the component level to the aircraft level were taken simultaneously. This was done to create a framework to understand if a TMS can support the intended operational and aircraft capabilities. Finally in Paper V, a top-down approach like the one taken in Paper I was employed to evaluate AI support in vehicle system design at the aircraft concept stage.

5.2. Key Results

The approaches used, methods created, and results obtained from these five papers were used to create a framework at the aircraft concept stage to support TM design. This framework is shown in Figure 14. The framework consists of creating a cross-functional TM design team, the other functional groups and external entities involved in TM design, and a timeline with stages for effective TM design. The timeline is supported by methods and tools for ECS design, evaluation, and detailed modelling and analysis. The timeline is a summary of the TM design timeline for the aircraft concept stage created by the cross-functional team in the study in Paper IV. Therefore, the cross-functional team defined the timeline that is presented in Figure 14.

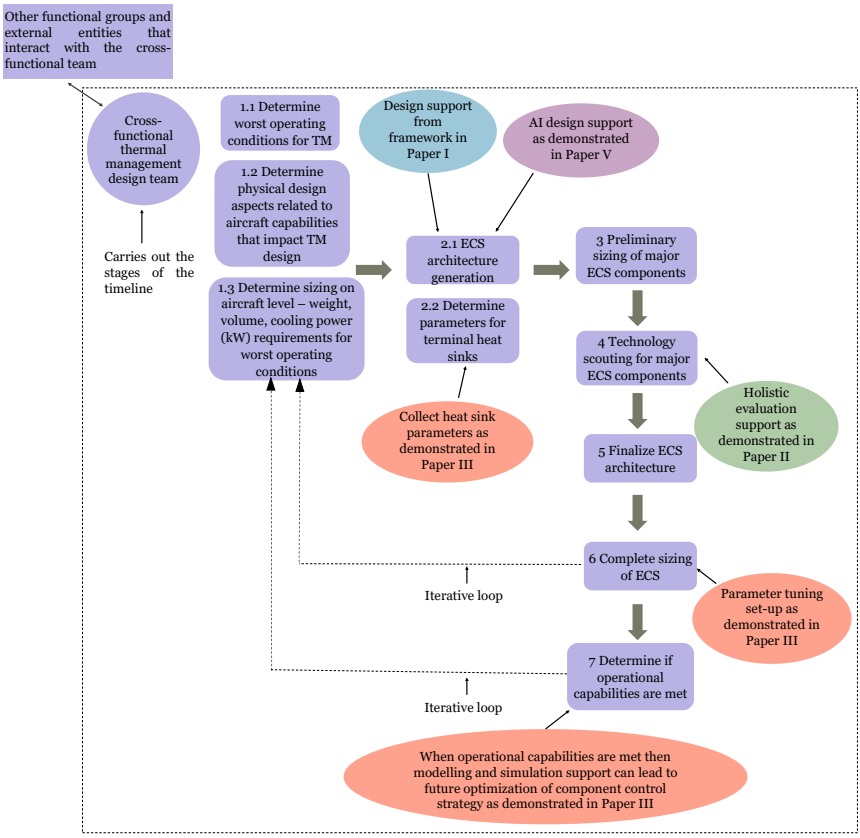


Figure 14 The framework demonstrates how effective thermal management (TM) design can be conducted at the concept stage at an aircraft developer

The cross-functional team for TM design is the same as that in Figure 9. The choice of experts constituting this team was based on data collected in the workshop described in Paper IV. The other functional groups and external entities that interact with the team are the same as shown in Figure 10. It is the cross-functional team that would conduct the activities in each stage of the timeline and interact with other functional groups and entities when required as detailed in Paper IV.

The first stage of the timeline would entail three steps 1.1, 1.2, and 1.3 that can be conducted simultaneously. Determining the worst operating conditions for TM can be done by the operational analysis expert in collaboration with the ECS and fuel system expert. The physical design aspects related to aircraft capabilities that impact TM design can be defined by the ECS and liaising experts. Dimensioning on an aircraft level for

weight, volume, and power (kW) requirements can be done by the liaising and overall aircraft design experts. When this information is available, the ECS architectures can be generated. Simultaneously, the parameters of all available heat sinks in the aircraft can be determined. This would include the mass flow rate, inlet temperature and pressure of the heat sink fluids. Architecture generation can be supported by the eight-step framework from Paper I as shown in Figure 6 here. It can also be supported by an AI tool trained on Saab-specific knowledge on ECS design. Terminal heat sink parameters for worst operating conditions can be defined as done in the parameter tuning study set-up in Paper III and shown in Figure 7 here. When the parameters of all available heat sinks are known, the preliminary sizing of major components can be done. This is so that the most effective terminal heat sink can be coupled to the heat exchangers for worst operating conditions.

The fourth stage entails ECS technology scouting. In this framework technology scouting is considered in a broader context. Aside from state-of-the-art commercial off the shelf equipment from vendors, technology scouting can also entail the aircraft developer look within the organization for technologies in existing aircraft. It could also include the organization developing their own technologies for a specific aircraft project in collaboration with vendors. The holistic approach to ECS evaluation in Paper II can be used to evaluate various technologies during the scouting phase as shown in Table 2.

When the major components are decided on, the ECS architecture can be finalized in the fifth stage followed by the complete sizing of the ECS in the sixth stage. Modelling and simulation studies for ECS architecture analysis at the sixth stage can be set up in a similar way to the simulation set-up in Paper III and as shown in Figure 8 here. The sizing of the ECS entails determining its maximum cooling capacity for the cooling power demand in the worst operating conditions. This determines the restrictions in the envelope and the length of use of mission-critical sensors in these conditions. Therefore, in the final stage if the cooling capacity falls below the required demand, then a second iteration commences at the second stage (2.1 and 2.2 in Figure 14) to meet operational capabilities. Finally, when the operational capabilities are met, system component modelling, and dynamic simulation and analysis as done in Paper

III can be conducted. This helps set up the method for ECS component control optimization. For example, control optimization of an electrical compressor could lead to less electrical power demand from the aircraft. Therefore, the framework consists of the appropriate experts that form the core cross-functional TM design team, and other functional groups and entities external to the organization that provide to or are impacted by TM design. It also consists of the iterative timeline for effective TM design defined by the team, and the use of supporting frameworks, methods and tools at suitable stages in the timeline. The framework demonstrates how effective TM design can be conducted at the concept stage at an aircraft developer.

6. Discussion

The answers to the research questions formulated in Chapter 1 are provided in this section. The answers are based on the key results presented in Chapter 5.

6.1. Answers to Research Questions

RQ1: How can effective TM design be conducted at the concept stage at an aircraft developer?
The framework pictorially presented in Figure 14 and described in Chapter 5 answer this question. The framework presented in this thesis consists of the appropriate experts that form the core cross-functional TM design team. It outlines the other functional groups and entities external to the organization that provide to or are impacted by TM design. It also consists of the iterative timeline for effective TM design defined by the team. The various stages of the timeline are carried out by the team with the support of various frameworks, methods and tools. The framework demonstrates how effective TM design can be conducted at the concept stage at an aircraft developer.

RQ2: How and when can control optimization of system components commence?
When the operational capabilities are met, system component modelling and dynamic simulation can be done in a similar way as demonstrated in Paper III. A parameter tuning study can be set up. It can be followed by detailed modelling and simulation of the system and its components. This helps set up the method for system component control optimization.

7. Conclusions

This chapter highlights the industrial and scientific contributions of the thesis. It also briefly describes future work that aligns with and builds on this thesis.

7.1. Industrial Contribution

The primary goal of this thesis was to influence thermal management design practice at the aircraft concept stage. The framework presented in this thesis demonstrates how effective TM design at the concept stage can be conducted at an aircraft developer like Saab. The framework was created with data collection through a cross-functional team workshop in Paper IV, a workshop in the form of a Turing test in Paper V, interviews in Paper II and Paper IV, document studies in Paper II, and group discussions in Paper IV. Therefore, the framework is based on industrial reality. It can be tested for application at the aircraft concept stage of Saab projects.

The various methods presented in the appended papers can be used for other applications at Saab. The workshop as a methodology in Paper IV can be applied at Saab to create frameworks for effective design of other basic aircraft functions. Papers I, II, III, and V used the ECS or an aircraft cooling system as a case study. However, the methods and frameworks created in these papers can also be applied to other aircraft subsystems. The framework in Paper I can be applied to generate architectures for aircraft vehicle systems other than the ECS at the concept stage. Similarly, the holistic evaluation approach presented in Paper II can be used to evaluate various vehicle system architectures. The parameter tuning study set-up can be applied to tuning studies of other vehicle systems. Finally, the method presented in Paper V can be applied to evaluate the usefulness of AI tools for design of other aircraft subsystems. Therefore, a second industrial contribution of this thesis is the creation of methods and frameworks that can

be used in a broader context that transcend their primary application in thermal management design.

7.2. Scientific Contribution

The primary goal of Paper IV was to demonstrate what is required for effective TM design at the aircraft concept stage. The framework presented in Paper IV can complement other methods in literature for TM architecture generation and TM modelling and simulation. It can do so because it presents the order in which information needed to populate these various methods is available, the dependencies between the information, and the various functional groups that need to be involved to obtain it.

One of the goals of Paper III was to enable an aircraft cooling system designer to optimize the control strategy of the VCS and its components for variable operation. System component modelling, and dynamic simulation and analysis presented in Paper III can serve as the preliminary method for component control optimization studies.

7.3. Future Work

Two studies to add new insights to the framework presented in this thesis have been planned. A brief description of the plan and the need for each of these studies is provided here.

The first study will focus on setting up a parameter tuning study for the aircraft fuel system architecture presented in Fig.6 of Paper IV. Design variables could include airframe skin thickness and material. Operational variables could include various missions covering a range of Mach numbers, altitudes, fuel mass flow rates, and amount of time spent in mission segments. This study is motivated by the shrinking capability of available terminal heat sinks due to contemporary aircraft design choices such as more use of composite materials for airframe skin. The study could result in a set-up for an optimization study of the use of fuel as a heat sink for variable operation.

The second study will focus on using the holistic evaluation approach from Paper II to firstly understand the emergent properties needed to

evaluate electrical technologies for vehicle systems such as actuation and ECS. This could include electro-hydrostatic actuators and electrically driven ECS. Then use those properties to compare these technologies to conventional technologies such as hydraulic actuators and bleed-air driven ECS. More electric technologies are becoming more common on aircraft. The advantages and disadvantages with these technologies need to be understood from a holistic perspective when compared to conventional technologies.

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Appended Papers

Papers I, II, III, IV, and V are appended to this thesis.

The papers have been removed for copyright reasons. For more details about these see:

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